

Internship proposal

Elastic surface fitting on implicit distance fields

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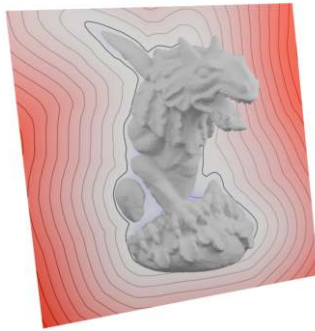
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The CRAFT team is an Inria project that aims at developing methods for designing and modeling deformable objects by combining geometry, physics, and numerical optimization. Our applications call for the setup and the resolution of inverse problems involving geometrical objects that needs to deform according to physical forces and material constraints.

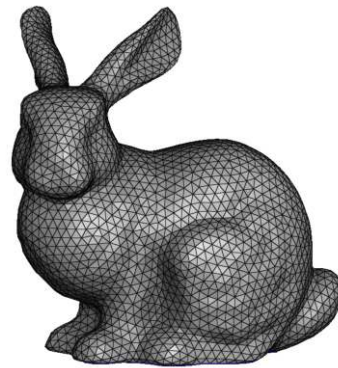
In this internship, we will focus on the deformation and evolution of an elastic object constrained to conform to a target surface. Such problems appear in the design of curved inflatable structures [10], surface reconstruction [4, 9] or remeshing algorithms (*shrinkwrap*) [3].

The (fixed) target will be represented by its implicit signed distance field. Recent advances in deep learning and neural representations allow to encode it as a neural network (Figure 1a), enabling automatic differentiation and quick evaluation of the distance and its gradient. On the other hand, the deformable object will be a polygonal mesh (Figure 1b) [2], the most widespread data structure used in this context. By considering the elastic properties of the material [7, 1], our goal will be to use the fitting energy as an external deformation force to study the properties of the object at mechanical equilibrium.

Special care will be given to the validity of the meshed surface, which has faces that can numerically generate, flip or overlap, such that the represented object does not make physical sense anymore. As such, the simulation of physical equilibrium will need to be modulated by a validity condition, which takes the form of regular remeshing (as in *shrinkwrapping* algorithms [? 3]), distortion energies [6, 8] or hard-constraints over the available degrees of freedom [4].



(a) Neural distance field of a gargoyle computing with the method of Coiffier and Béthune [5].



(b) Triangle mesh of the *Stanford Bunny*.

Required skills

The intern will have to be comfortable with either python or C++ for implementation. They should have some knowledge about geometry processing data structure and algorithms, as well as numerical optimization algorithms.

Practical information

The internship will take place in the CRAFT research team in the Inria center of the Grenoble Alpes University (655 Avenue de l'Europe, 38330 Montbonnot-Saint-Martin, France).

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